

CIDR Notation and Subnet Masks

Section 1: Foundations: What Are Subnet Masks and CIDR?

1.1 The Structure of an IP Address

An IPv4 address is a 32-bit number. To make it easier for humans to read, it is written as four groups of numbers separated by dots, for example 192.168.1.1. Each group is called an octet because it contains exactly 8 bits. Four octets multiplied by 8 bits each gives us 32 bits in total.

Every IP address has two parts. The first part identifies the network, which is like identifying the street or neighborhood. The second part identifies the specific device on that network, which is like identifying the individual house number on that street. The question that subnet masks and CIDR notation both answer is: where does the network part end and the device part begin?

1.2 What Is a Subnet Mask?

A subnet mask is a 32-bit number written in the same dotted decimal format as an IP address. It uses a sequence of 1-bits followed by a sequence of 0-bits. The 1-bits mark the network portion of the address, and the 0-bits mark the host portion. There are no gaps or alternating patterns. Instead, it is a continuous block of 1-bits on the left, followed by a continuous block of 0-bits on the right.

For example, the subnet mask 255.255.255.0 in binary looks like this:

255	255	255	0
11111111	11111111	11111111	00000000
<-- Network ---> <-Host->			

The twenty-four 1-bits on the left tell the network device: the first 24 bits of any IP address using this mask belong to the network. The eight 0-bits on the right say: the last 8 bits identify the individual device.

1.3 What Is CIDR Notation?

CIDR stands for Classless Inter-Domain Routing. It was introduced in 1993 to replace the older classful system of networking, which divided all addresses rigidly into Class A, Class B, and Class C blocks. CIDR allows networks to be any size rather than being locked into predefined classes.

In CIDR notation, instead of writing out the full subnet mask, you simply write a forward slash followed by the number of 1-bits in that mask. For example:

CIDR Notation Examples

255.255.255.0 is written as **/24** because there are 24 consecutive 1-bits

255.255.0.0 is written as **/16** because there are 16 consecutive 1-bits

255.0.0.0 is written as **/8** because there are 8 consecutive 1-bits

255.255.255.192 is written as **/26** because there are 26 consecutive 1-bits

CIDR notation is always attached to an IP address using a slash. For example, 192.168.1.0/24 means the IP address 192.168.1.0 with a subnet mask that has 24 network bits. This tells you the network address, the size of the network, and by extension how many devices it can hold, all in a single compact expression.

Section 2: Binary Reference Table

Before we begin the conversions, it is helpful to have a reference for the binary values of the numbers that appear most commonly in subnet masks. These are the only values you will

ever see in the last octet of a subnet mask, because a valid subnet mask must always be a continuous block of 1-bits.

Decimal Value	Binary Representation	Number of 1-bits	Meaning
255	11111111	8	All 8 bits are network bits
254	11111110	7	7 network bits, 1 host bit
252	11111100	6	6 network bits, 2 host bits
248	11111000	5	5 network bits, 3 host bits
240	11110000	4	4 network bits, 4 host bits
224	11100000	3	3 network bits, 5 host bits
192	11000000	2	2 network bits, 6 host bits
128	10000000	1	1 network bit, 7 host bits
0	00000000	0	All 8 bits are host bits

Note:

The pattern of valid last-octet values follows powers of 2. Each value is exactly 128, 64, 32, 16, 8, 4 or 2 less than the one before it. This is because each additional 1-bit in the binary number adds the next power of 2 (128, 64, 32, 16, 8, 4, 2, 1) from the left.

Section 3: Converting Subnet Mask to CIDR Notation

3.1 The Method

To convert a subnet mask to CIDR notation, follow these steps:

- **Step 1:** Convert each of the four octets from decimal to binary, giving you 8 bits for each octet.
- **Step 2:** Write all four octets side by side to give you a 32-bit binary number.
- **Step 3:** Count every 1-bit from left to right. Stop counting when you hit the first 0-bit. The total count is your CIDR number.

3.2 Examples

1: 255.0.0.0

Convert each octet to binary:

Octet 1: 255 = 11111111 (8 ones)
Octet 2: 0 = 00000000 (0 ones)
Octet 3: 0 = 00000000 (0 ones)
Octet 4: 0 = 00000000 (0 ones)
Full 32-bit binary: 11111111.00000000.00000000.00000000
Count of 1-bits: $8 + 0 + 0 + 0 = 8$

Answer: /8

2: 255.255.0.0

Octet 1: 255 = 11111111 (8 ones)
Octet 2: 255 = 11111111 (8 ones)
Octet 3: 0 = 00000000 (0 ones)
Octet 4: 0 = 00000000 (0 ones)
Full 32-bit binary: 11111111.11111111.00000000.00000000
Count of 1-bits: $8 + 8 + 0 + 0 = 16$

Answer: /16

3: 255.255.255.0

Octet 1: 255 = 11111111 (8 ones)
Octet 2: 255 = 11111111 (8 ones)
Octet 3: 255 = 11111111 (8 ones)
Octet 4: 0 = 00000000 (0 ones)
Full 32-bit binary: 11111111.11111111.11111111.00000000
Count of 1-bits: $8 + 8 + 8 + 0 = 24$
Answer: /24

Section 4: Converting CIDR Notation to Subnet Mask

4.1 The Method

Converting CIDR notation to a subnet mask is essentially the reverse of Section 3. The steps are:

- **Step 1:** Take the CIDR number. Write exactly that many 1-bits starting from the left.
- **Step 2:** Fill all remaining bit positions with 0-bits until you have exactly 32 bits in total.
- **Step 3:** Divide the 32 bits into four groups of 8 bits each (four octets).
- **Step 4:** Convert each octet from binary to decimal using the place value table.
- **Step 5:** Write the four decimal numbers separated by dots. That is your subnet mask.

4.2 Examples

1: /8

Write 8 ones, then 24 zeros:

11111111.00000000.00000000.00000000

Convert each octet:

Octet 1: 11111111 = $128+64+32+16+8+4+2+1 = 255$

Octet 2: 00000000 = 0

Octet 3: 00000000 = 0

Octet 4: 00000000 = 0

Answer: 255.0.0.0

2: /16

Write 16 ones, then 16 zeros:

11111111.11111111.00000000.00000000

Convert each octet:

Octet 1: 11111111 = 255

Octet 2: 11111111 = 255

Octet 3: 00000000 = 0

Octet 4: 00000000 = 0

Answer: 255.255.0.0

Example 3: /24

Write 24 ones, then 8 zeros:

11111111.11111111.11111111.00000000

Octet 1: 255 Octet 2: 255 Octet 3: 255 Octet 4: 0

Answer: 255.255.255.0

Section 5: Calculating Usable Host Capacity

5.1 The Concept of Host Bits

Every IP address has 32 bits in total. The CIDR number tells us how many of those bits are used for the network. Whatever is left over belongs to the host portion, meaning those bits are used to identify individual devices on the network.

Host Bits Formula

Host bits = 32 minus the CIDR number

For example, with a /24 network: host bits = $32 - 24 = 8$. Those 8 bits can represent devices.

5.2 Calculating Total Addresses

Once you know how many bits are available for hosts, you calculate the total number of addresses those bits can produce by raising 2 to the power of the number of host bits.

The reason we use powers of 2 is because each bit has exactly two possible states: it can be either 0 or 1. Each additional bit doubles the number of combinations available. With 1 bit you can have 2 combinations (0 or 1). With 2 bits you can have 4 combinations (00, 01, 10, 11). With 8 bits you can have 256 combinations.

Total Addresses Formula

Total addresses = 2 raised to the power of host bits

Written mathematically: $\text{Total} = 2^{(\text{host bits})}$

5.3 Why We Subtract 2 for Usable Hosts

Not every address in a network block can be assigned to a device. Two addresses are always reserved:

- **The Network Address:** This is the very first address in the block. It is formed by setting all host bits to 0. This address identifies the network itself and cannot be assigned to any device.
- **The Broadcast Address:** This is the very last address in the block. It is formed by setting all host bits to 1. When data is sent to the broadcast address, it is delivered to every device on the network simultaneously. It cannot be assigned to a single device.

Usable Hosts Formula

$$\text{Usable hosts} = 2^{(\text{host bits})} - 2$$

We subtract 1 for the network address and 1 for the broadcast address, giving us minus 2 in total.

5.4 Examples

1: /24

Step 1 – Find host bits:

$$\text{Host bits} = 32 - 24 = 8$$

Step 2 – Calculate total addresses:

$$\text{Total} = 2^8$$

$$2^8 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256$$

Step 3 – Subtract reserved addresses:

$$\text{Usable} = 256 - 2 = 254$$

Answer: 254 usable host addresses

2: /25

$$\text{Step 1 – Host bits} = 32 - 25 = 7$$

$$\text{Step 2 – Total} = 2^7 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 128$$

$$\text{Step 3 – Usable} = 128 - 2 = 126$$

Answer: 126 usable host addresses

3: /26

$$\text{Step 1 – Host bits} = 32 - 26 = 6$$

$$\text{Step 2 – Total} = 2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$$

Step 3 – Usable = $64 - 2 = 62$

Answer: 62 usable host addresses

Section 6: Full Address Range Breakdown

For this section, we are working with the specific network address **192.168.10.0/26**. We will answer five questions about it in full detail.

Question 1: What is the subnet mask?

The CIDR notation is /26, meaning we need 26 consecutive 1-bits followed by 6 zero-bits.

Write 26 ones then 6 zeros:

11111111.11111111.11111111.11000000

The first three octets are fully 1-bits:

Octet 1: 11111111 = 255

Octet 2: 11111111 = 255

Octet 3: 11111111 = 255

The last octet has the 25th and 26th bits active:

Octet 4: 11000000

Place values: 128 64 32 16 8 4 2 1

Binary: 1 1 0 0 0 0 0 0

Active values: $128 + 64 = 192$

Answer: 255.255.255.192

Question 2: How many usable hosts are available?

Step 1 – Find host bits:

Host bits = $32 - 26 = 6$

Step 2 – Calculate total addresses:

Total = $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$

Step 3 – Subtract reserved addresses:

Usable = $64 - 2 = 62$

Answer: 62 usable host addresses

Question 3: What is the network address?

The network address is the first address in the block. It is found by taking the given IP address and setting all host bits to 0.

In this case, the given address is 192.168.10.0. The last 6 bits are the host bits (because /26 means 26 network bits, leaving 6 host bits in the last octet).

Given IP address: 192.168.10.0

Last octet in binary: 00000000

For a /26, the last 6 bits are host bits.

Network bits in last octet: first 2 bits = 00

Host bits in last octet: last 6 bits = 000000

Setting all host bits to 0:

Last octet binary: 00 | 000000 = 00000000 = 0

Network address: 192.168.10.0

Answer: 192.168.10.0

Note

In this example, the given address was already the network address because the last octet was 0. In other cases, you may need to zero out the host bits to find the actual network address.

Question 4: What is the broadcast address?

The broadcast address is the last address in the block. It is found by taking the network address and setting all host bits to 1.

Network address: 192.168.10.0

Last octet in binary: 00000000

For a /26, the last 6 bits are host bits.

Network bits in last octet (first 2): 00

Host bits in last octet (last 6): 000000

Setting all host bits to 1:

Last octet binary: 00 | 111111

Convert 00111111 to decimal:

Place values:	128	64	32	16	8	4	2	1
Binary:	0	0	1	1	1	1	1	1
Active values:			32	+ 16	+ 8	+ 4	+ 2	+ 1
	= 32 + 16 + 8 + 4 + 2 + 1 = 63							

Broadcast address: 192.168.10.63

Answer: 192.168.10.63

Question 5: What is the usable host range?

The usable host range is every address between the network address and the broadcast address, excluding both ends. The network address and broadcast address are reserved and cannot be assigned to any device.

Network address:	192.168.10.0	<- Reserved, identifies the network
First usable host:	192.168.10.1	<- First device can use this
	192.168.10.2	
	192.168.10.3	
	...	
	192.168.10.61	
Last usable host:	192.168.10.62	<- Last device can use this
Broadcast address:	192.168.10.63	<- Reserved, for broadcast traffic

Total usable addresses: 62 (from .1 to .62)

Answer: 192.168.10.1 to 192.168.10.62

6.1 Summary

Question	Answer
Subnet mask	255.255.255.192
Usable hosts	62 devices
Network address	192.168.10.0 (all host bits = 0)
Broadcast address	192.168.10.63 (all host bits = 1)

Question	Answer
Usable host range	192.168.10.1 to 192.168.10.62

Section 7: Key Takeaways

These are the core principles from everything covered in this document:

- **CIDR and subnet masks are two ways of expressing the same thing.** They both tell you where the network portion of an IP address ends and the host portion begins. CIDR is simply the shorter, more modern way to write it.
- **The CIDR number equals the count of 1-bits in the subnet mask.** Count the 1-bits from left to right and you have your CIDR number. Write that many 1-bits and fill the rest with 0-bits and you have your subnet mask.
- **Host bits determine network capacity.** Subtract the CIDR number from 32 to find the host bits. Raise 2 to the power of those host bits to get total addresses. Subtract 2 for the reserved network and broadcast addresses to get usable hosts.
- **Every network has a network address and a broadcast address.** The network address has all host bits set to 0. The broadcast address has all host bits set to 1. Neither can be assigned to a device.
- **Each increase of 1 in the CIDR number halves the available host space.** This is because one more bit is taken from the host side. Losing one bit always cuts the number of combinations in half.